Chapter 1

Questions on

Lesson Three

Ohm's Law for the Closed Circuit

Interactive test



The questions signed by ✻ are answered in detail. ● Understand ○ Apply ∴ Higher Order Thinking Skills

Important guidelines

Ohm's law for the closed circuit

$$V_B = V + Ir = IR + Ir = I\,(R + r)$$

To determine the electric current intensity (I) in a closed electric circuit:

$$I = \frac{V_B}{R + r} = \frac{V_B - V}{r}$$

- In the case of the absence of electric current (I = 0) then: $V = V_B$

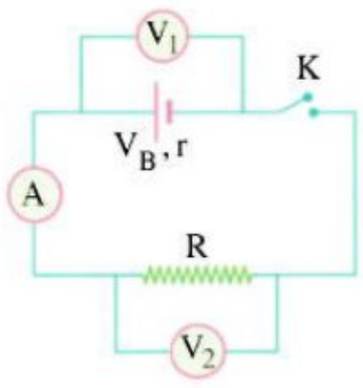
(1) In the following electric circuit if switch K is:

Opened

$I = 0$

$V_2 = 0$

$V_1 = V_B$



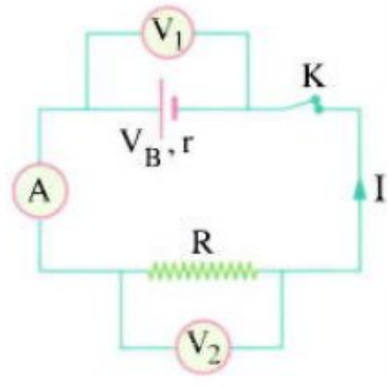
Closed

$$I = \frac{V_B}{R + r}$$

$$= \frac{V_B - V_1}{r} = \frac{V_2}{R}$$

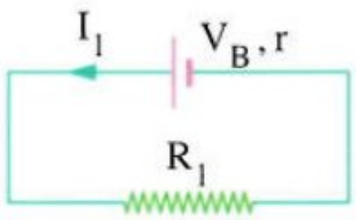
$V_2 = IR$

$V_1 = V_B - Ir$

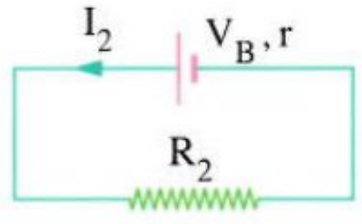


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(2) When the external resistor R_1, which carries electric current of intensity I_1 is replaced by another R_2, the passing electric current intensity in the circuit changes to I_2 when using the same battery:



$V_B = I_1 (R_1 + r)$

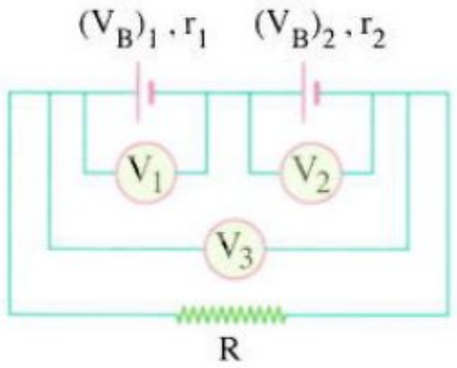


$V_B = I_2 (R_2 + r)$

Thus the two equations are solved algebraically to find the unknown values.

(3) When two electric cells are connected as follows:

①



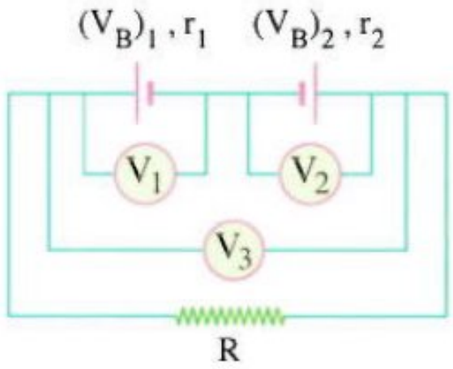
$$I = \frac{(V_B)_1 + (V_B)_2}{R + r_1 + r_2}$$

$V_1 = (V_B)_1 - Ir_1$ (Discharging)

$V_2 = (V_B)_2 - Ir_2$ (Discharging)

$V_3 = V_1 + V_2 = IR$

②



$(V_B)_1 > (V_B)_2$

$$I = \frac{(V_B)_1 - (V_B)_2}{R + r_1 + r_2}$$

$V_1 = (V_B)_1 - Ir_1$ (Discharging)

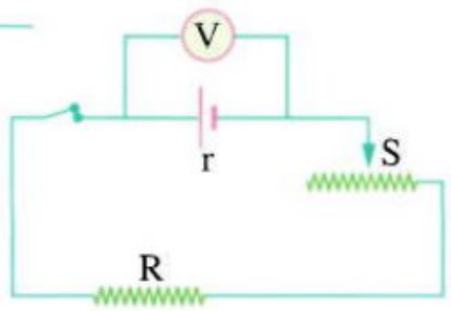
$V_2 = (V_B)_2 + Ir_2$ (Charging)

$V_3 = V_1 - V_2 = IR$

First Multiple choice questions

1. If the emf of an electric source whose internal resistance is r equals 8 V, the potential difference between its terminals during the flow of electric current in its circuit ..... .
   - (a) equals 8 V
   - (b) is less than 8 V
   - (c) is more than 8 V
   - (d) the answer can't be determined without knowing the value of r

2. In the opposite electric circuit, when the variable resistor (S) increases, the reading of the voltmeter ..... .
   - (a) increases
   - (b) decreases until it vanishes
   - (c) remains constant
   - (d) decreases but doesn't vanish



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3 ✱ If a battery of emf 6 V is connected to a 10 Ω resistor, an electric current of intensity 0.5 A passes in the circuit, so the internal resistance of the battery equals ..... .

(a) 0.5 Ω (b) 1 Ω (c) 1.5 Ω (d) 2 Ω

4 ✱ Three resistors 3 Ω, 6 Ω and 4 Ω are connected in series to a battery of emf 30 V and internal resistance 2 Ω, so:

(i) The total resistance of the circuit equals ..... .

(a) 11 Ω (b) 13 Ω (c) 15 Ω (d) 19 Ω

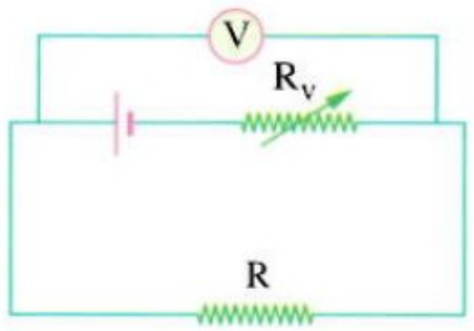
(ii) The intensity of the electric current in the circuit equals ..... .

(a) 0.5 A (b) 1 A (c) 1.5 A (d) 2 A

(iii) The potential difference across the terminals of the 6 Ω resistor equals ..... .

(a) 6 V (b) 8 V (c) 12 V (d) 18 V

5 In the opposite electric circuit, when R_v decreases, the voltmeter reading (V) ..... .

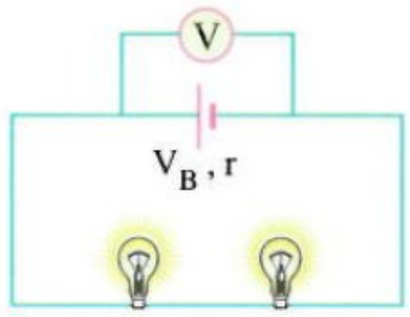


(a) increases
(b) decreases till it vanishes
(c) remains constant
(d) decreases but doesn't vanish

6 The intensity of the current that flows through a battery after closing its external circuit is inversely proportional to ..... .

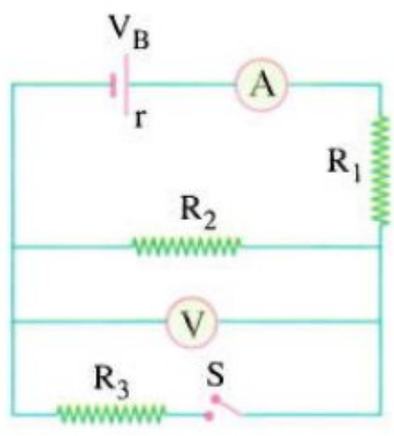
(a) the emf of the battery (b) the internal resistance of the battery
(c) the external equivalent resistance (d) the total resistance of the circuit

7 In the opposite electric circuit, if the filament of one of the lightbulbs burns out, the voltmeter reading ..... .



(a) increases
(b) decreases but does not equal zero
(c) remains constant
(d) becomes zero

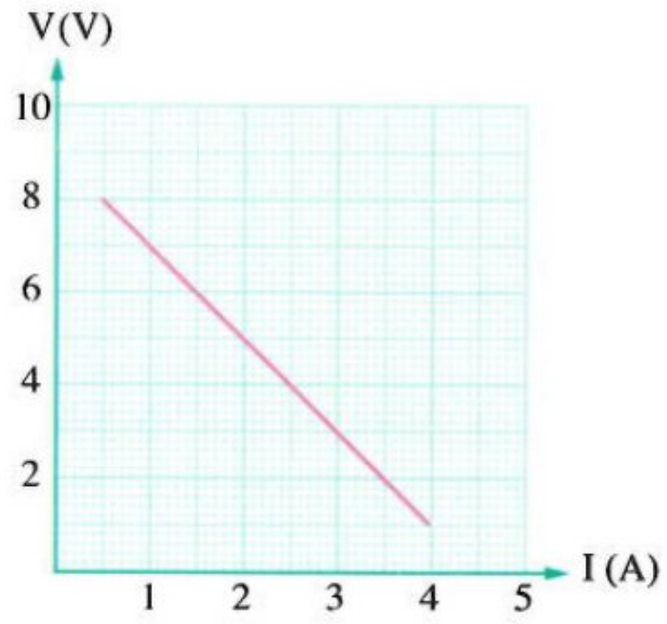
8 In the opposite electric circuit when switch S is closed, then the readings of the voltmeter and the ammeter ..... , ... respectively.



(a) decreases, decreases
(b) decreases, increases
(c) increases, decreases
(d) increases, increases

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9 ✳ The opposite graph represents the relation between the potential difference across the terminals of a battery (V) and the current intensity (I) that passes through it, then:



(i) The internal resistance of the battery is ..... .

(a) 1 Ω (b) 2 Ω

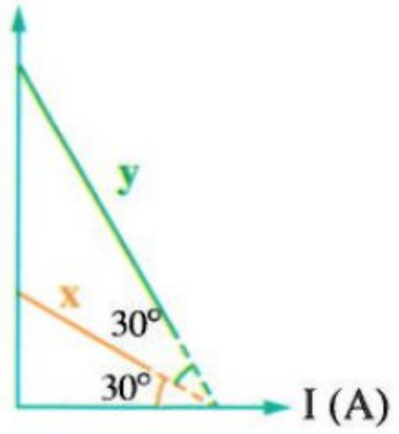
(c) 0.45 Ω (d) 0.9 Ω

(ii) The emf of the battery is ..... .

(a) 7 V (b) 7.5 V

(c) 8 V (d) 9 V

10 The opposite graph represents the variation of the potential difference between the two poles of two electric cells (x), (y) with respect to the electric current intensity that passes in the circuit of each of them, then the ratio between their internal resistances $\left(\frac{r_x}{r_y}\right)$ is ..... .



(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{\frac{3}{2}}$

11 Figure (1) represents the graph of voltmeter reading (V) versus ammeter reading (A) when changing the value of the variable resistor (R) in the electrical circuit shown in figure (2):

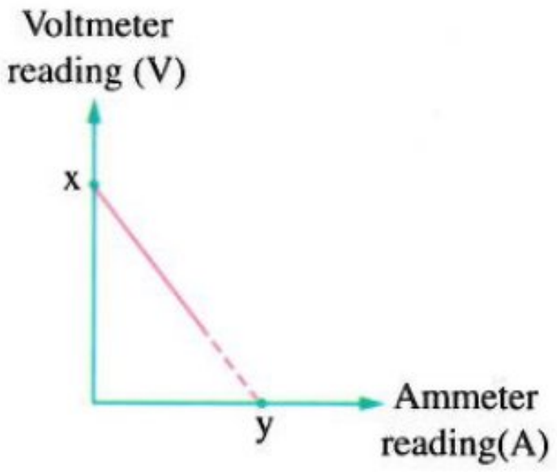


Figure (1)

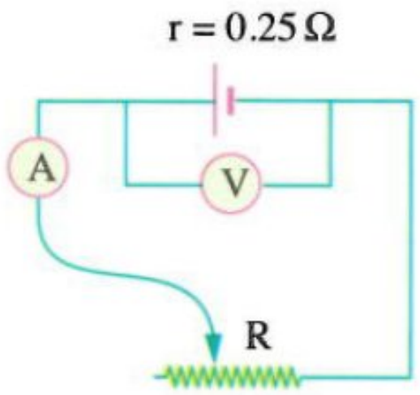


Figure (2)

If the internal resistance of the battery is 0.25 Ω, which of the following values represent two possible values for the two points (x) and (y) on the graph in figure (1)?

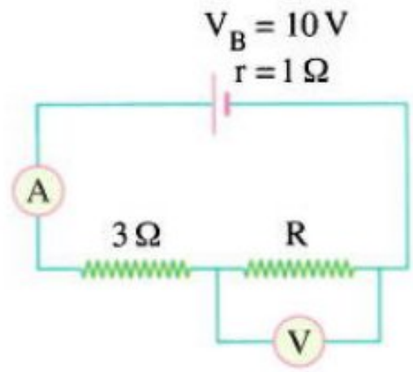
	x	y
(a)	8 V	8 A
(b)	8 V	2 A
(c)	2 V	8 A
(d)	2 V	4 A

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12 In the electric circuit shown in the figure, if the reading of the ammeter is 1 A, the reading of the voltmeter is ..... .



(a) 3 V (b) 6 V
(c) 7 V (d) 9 V

13 ✱ A metal wire of length of 30 m, cross-sectional area 0.3 cm^2 and material resistivity 5×10^{-7} Ω.m is connected in series to a resistor of 8.5 Ω and a battery whose emf is 18 V and its internal resistance is 1 Ω. So, the electric current intensity passing in the circuit equals ..... .

(a) 0.9 A (b) 1.8 A (c) 3.6 A (d) 4.2 A

14 ✱ A battery that has emf of 6 V and internal resistance of 1 Ω is connected with an ammeter of negligible resistance, a constant resistance R and a rheostat together in series and when the slider is adjusted at the beginning of the rheostat, an electric current of 0.6 A passes in the circuit and when the slider is moved to the end of the rheostat, the current that passes in the circuit becomes 0.1 A, so:

(i) The resistance of R equals ..... .

(a) 3 Ω (b) 6 Ω (c) 9 Ω (d) 12 Ω

(ii) The maximum resistance of the rheostat equals ..... .

(a) 15 Ω (b) 27 Ω (c) 38 Ω (d) 50 Ω

15 ✱ Two resistors, $R_1 = 6$ Ω and $R_2 = 4$ Ω, are connected together in parallel between the terminals of an electric source that has emf of 6 V and internal resistance of 0.1 Ω, so:

(i) The intensity of the passing electric current in the circuit equals ..... .

(a) 1.3 A (b) 2.4 A (c) 3.6 A (d) 4 A

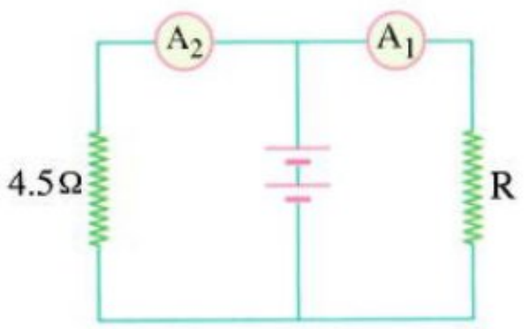
(ii) The consumed electric power in the circuit (inside and outside the source) equals ..... .

(a) 14.4 W (b) 15 W (c) 18.3 W (d) 21.9 W

(iii) The electric power that is consumed in R_1 equals ..... .

(a) 12.2 W (b) 9.3 W (c) 6.1 W (d) 5.53 W

16 ✱ In the opposite circuit: If the reading of ammeter $A_1 = 1$ A while the reading of ammeter $A_2 = 2$ A and the internal resistance of the battery (r) = 1 Ω, so:



(i) The resistance R equals ..... .

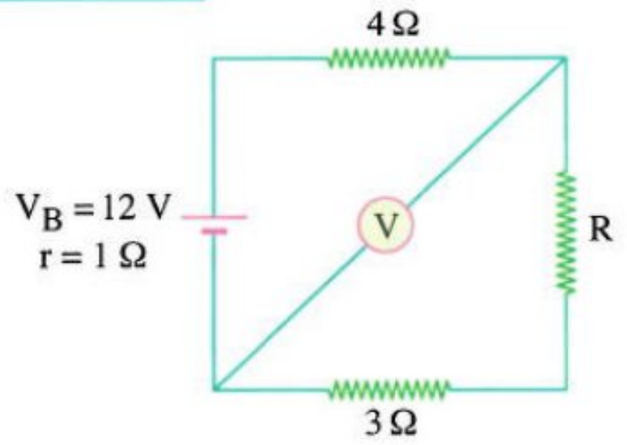
(a) 2 Ω (b) 4.5 Ω
(c) 9 Ω (d) 13 Ω

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(ii) The emf of the battery equals ..... .

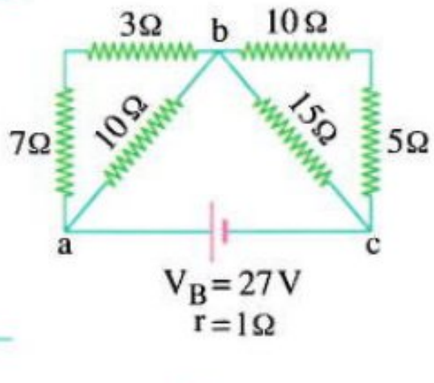
(a) 14 V (b) 12 V (c) 9 V (d) 8 V

17 In the opposite electric circuit, if the voltmeter reading is 6 V, the resistance R equals ..... .



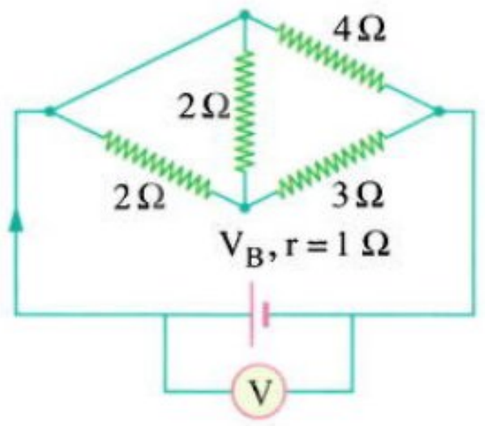
(a) 2 Ω
(b) 3 Ω
(c) 4.8 Ω
(d) 6 Ω

18 ✳ In the circuit shown in the figure, the potential difference between b and c equals ..... .



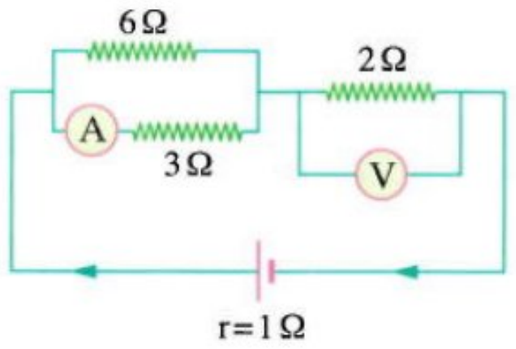
(a) 5 V (b) 10 V
(c) 15 V (d) 20 V

19 In the opposite electrical circuit, if the voltmeter reading is 8 V, then the electromotive force (V_B) of the battery equals ..... .



(a) 8 V
(b) 10 V
(c) 12 V
(d) 14 V

20 In the opposite electric circuit, if the current that passes in the 6 Ω resistor equals 1 A, then:



(i) The reading of the ammeter (A) equals ..... .

(a) 3.4 A (b) 3 A
(c) 2.8 A (d) 2 A

(ii) The reading of the voltmeter (V) equals ..... .

(a) 1 V (b) 3 V (c) 6 V (d) 9 V

(iii) The emf of the battery equals ..... .

(a) 11 V (b) 13 V (c) 15 V (d) 19 V

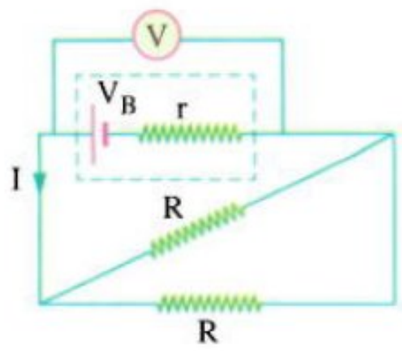
21 ✳ Three resistors of 8 Ω, 6 Ω and 16 Ω are connected together in a definite way and with an electric source of internal resistance 1.2 Ω. When the circuit is closed, the potential differences across the resistors become 2 V, 6 V and 4 V respectively, so the emf of the source equals ..... .

(a) 6 V (b) 7.5 V (c) 9 V (d) 12 V

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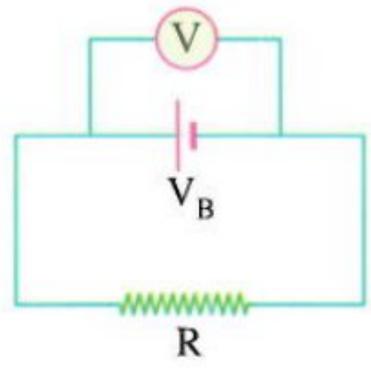
22 In the opposite electric circuit, the reading of the voltmeter equals .....



(a) $V_B - 2\,Ir$ (b) $V_B + Ir$

(c) $\frac{IR}{2}$ (d) $2\,IR$

23 In the opposite circuit, if the internal resistance of the battery is $\frac{1}{4}$ R, then the voltmeter reading equals ..... .

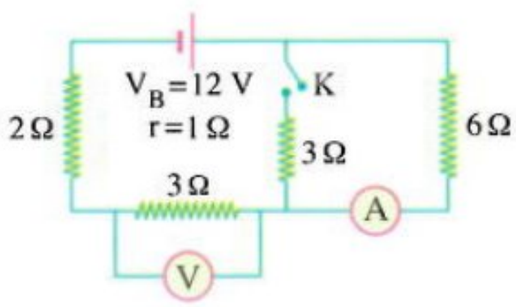


(a) $\frac{2}{3}\,V_B$ (b) $\frac{5}{4}\,V_B$

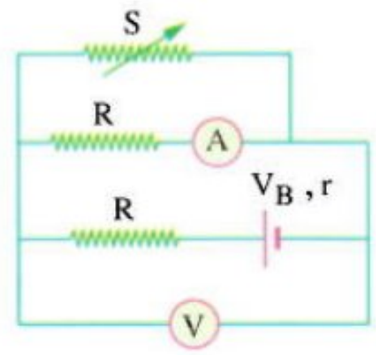
(c) $\frac{1}{5}\,V_B$ (d) $\frac{4}{5}\,V_B$

24 In the shown electric circuit, when the switch K is closed, then the ammeter (A) and voltmeter (V) readings are ..... .

	The ammeter reading (A)	The voltmeter reading (V)
(a)	increases	decreases
(b)	decreases	increases
(c)	increases	increases
(d)	decreases	decreases



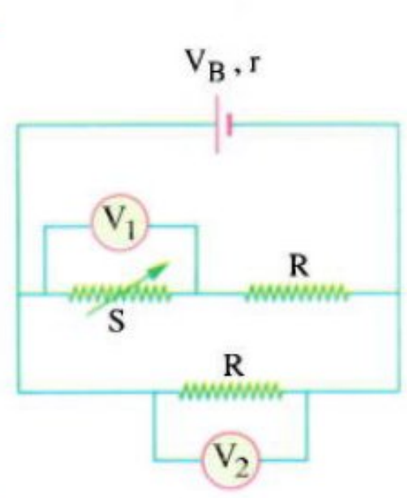
25 What happens to the readings of both ammeter and voltmeter respectively when increasing the variable resistance (S) in the opposite electrical circuit?



(a) Increases, increases. (b) Increases, decreases.

(c) Decreases, increases. (d) Decreases, decreases.

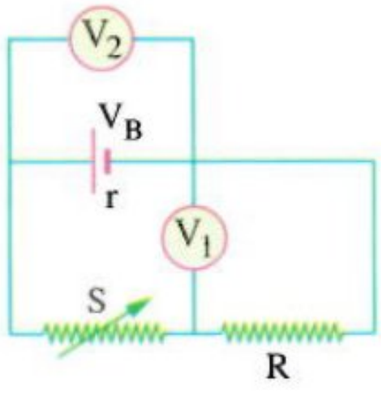
26 The opposite diagram represents a closed electric circuit, so when the variable resistance (S) is increased, the readings of the two voltmeters V_1 and V_2 are ....., ... respectively.



(a) increases, increases (b) decreases, decreases

(c) increases, decreases (d) decreases, increases

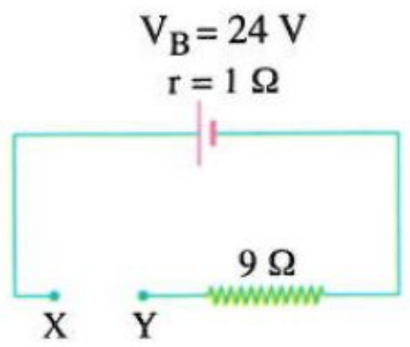
27 In the shown electric circuit, if the value of the variable resistance (S) is decreased, which of the following ratios decreases?



(a) $\frac{V_1}{V_B}$ (b) $\frac{V_2}{V_1}$ (c) $\frac{V_B}{V_2}$ (d) $\frac{V_1}{V_2}$

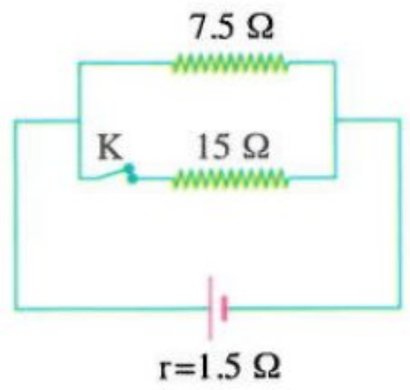
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28 If you have a group of resistors each of value 8 Ω, what is the number of these resistors and how can you connect them with each other between the two points X and Y in order to get a current intensity of 2 A passing in the circuit?



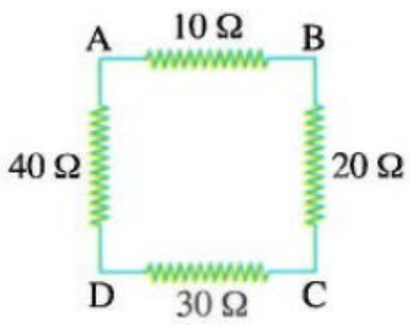
(a) 2 resistors, in series.
(b) 3 resistors, in series.
(c) 4 resistors, in parallel.
(d) 6 resistors, in parallel.

29 In the opposite electric circuit the current intensity that passes in the circuit in the case of closing the switch K is greater than the current intensity that passes in the case of opening the switch K by 0.5 A. So, the emf of the battery equals ..... .



(a) 8.2 V
(b) 9.4 V
(c) 10.3 V
(d) 11.7 V

30 ✳ The opposite figure shows four resistors which are connected in the form of a square ABCD:



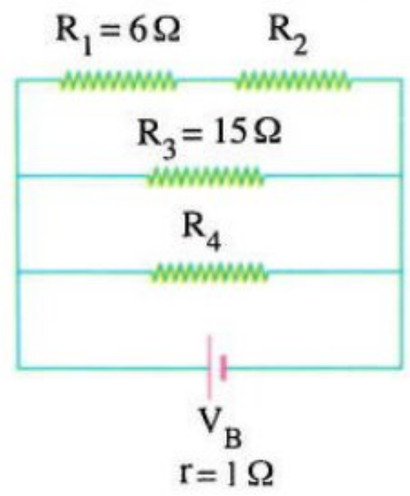
(i) So, the two points which have to be connected to a battery whose internal resistance is 1 Ω to let a current of 0.25 A pass in all resistors are ..... .

(a) A, B
(b) B, D
(c) C, D
(d) A, C

(ii) The emf of the battery which is connected in (i) equals ..... .

(a) 13 V
(b) 26 V
(c) 35 V
(d) 40 V

31 In the opposite figure an electric circuit consists of a battery V_B whose internal resistance is 1 Ω. The battery is connected to four resistors R_1, R_2, R_3 and R_4 that are carrying 0.3 A, 0.3 A, 0.4 A and 0.2 A respectively, so:



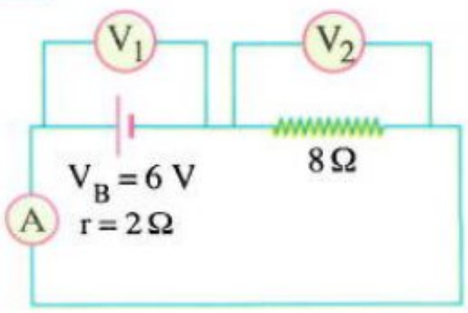
(i) The emf of the source V_B equals ..... .

(a) 5.4 V
(b) 6.9 V
(c) 7.13 V
(d) 9.51 V

(ii) The total resistance of the circuit equals ..... .

(a) 30 Ω
(b) 14 Ω
(c) 7.67 Ω
(d) 13.34 Ω

32 ✳ In the opposite electric circuit, the readings of the voltmeters V_1 and V_2 are ..... , ... respectively.

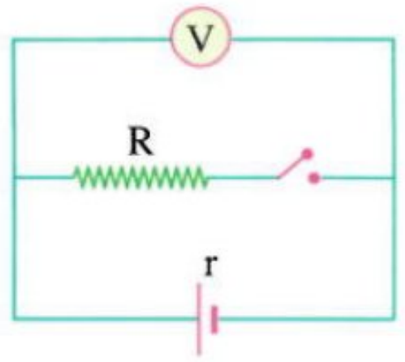


(a) 2.25 V, 5.5 V
(b) 4.8 V, 4.8 V
(c) 2.25 V, 2.25 V
(d) 4.8 V, 2.4 V

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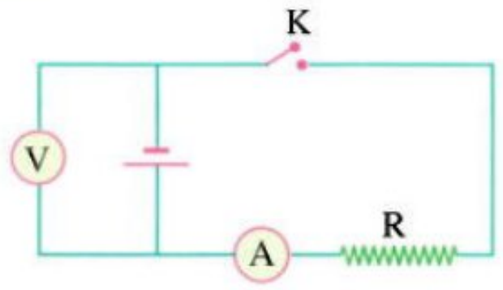
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33 A resistor (R) is connected to a battery of internal resistance r and a switch, as shown in the opposite electrical circuit. If the voltmeter reading is V_o when the switch is open, and 0.8 V_o when the switch is closed, then the ratio $\left(\frac{R}{r}\right)$ equals ..... .



(a) 1 (b) 2 (c) 4 (d) 8

34 ✳ In the opposite electric circuit, the reading of the voltmeter is 12 V when the switch K is opened and when the switch K is closed, its reading becomes 9 V where the reading of the ammeter is 1.5 A, so:



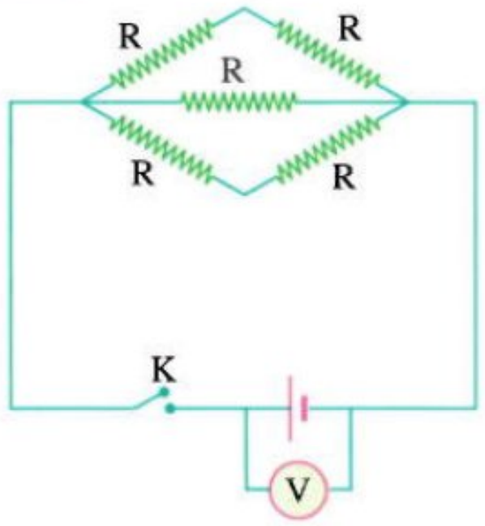
(i) The internal resistance of the battery equals ..... .

(a) 0.5 Ω (b) 1 Ω (c) 1.5 Ω (d) 2 Ω

(ii) The resistance of R equals ..... .

(a) 2 Ω (b) 4 Ω (c) 6 Ω (d) 8 Ω

35 ✳ In the opposite electric circuit, if the internal resistance of the battery is 2 Ω and the reading of the voltmeter is 12 V when the switch (K) is opened and it becomes 10 V when the switch (K) is closed, so:



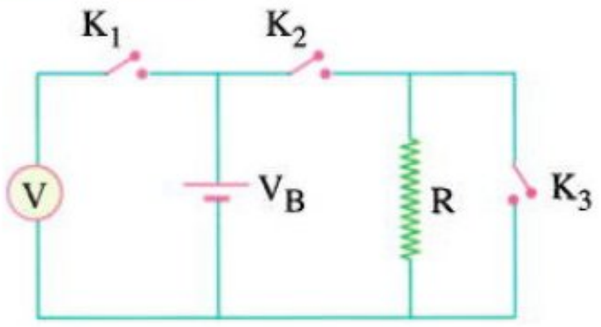
(i) The electric current intensity that passes in the circuit when the switch (K) is closed equals ..... .

(a) 4 A (b) 3 A (c) 2 A (d) 1 A

(ii) The resistance of R equals ..... .

(a) 5 Ω (b) 10 Ω (c) 15 Ω (d) 20 Ω

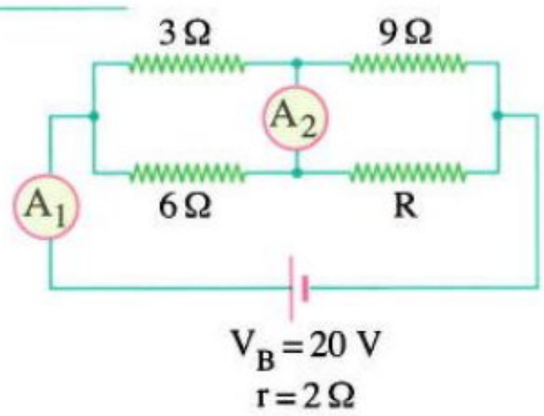
36 In the opposite electric circuit, a voltmeter is connected across a battery of emf (V_B) that has an internal resistance r, so the voltmeter gives a reading equal to zero at closing ..... .



(a) switch K_1 only (b) both switches K_1 and K_2 only

(c) both switches K_1 and K_3 only (d) all three switches K_1, K_2 and K_3

37 ✳ In the opposite electric circuit, if the pointer of the ammeter (A_2) is at zero, then the reading of the ammeter (A_1) is ..... .

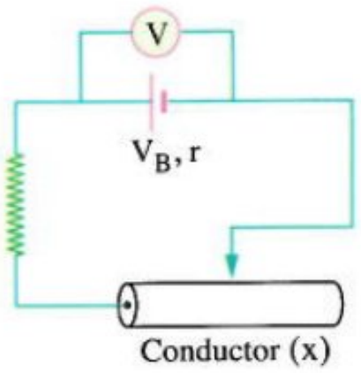


(a) 3.5 A (b) 2.5 A

(c) 2 A (d) 1.5 A

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38 An electric source is connected to a conductor (x) of uniform cross-section using a slider as shown in the opposite circuit, so what happens to the voltmeter reading (V) when increasing the segment length of conductor (x) that carries electric current?

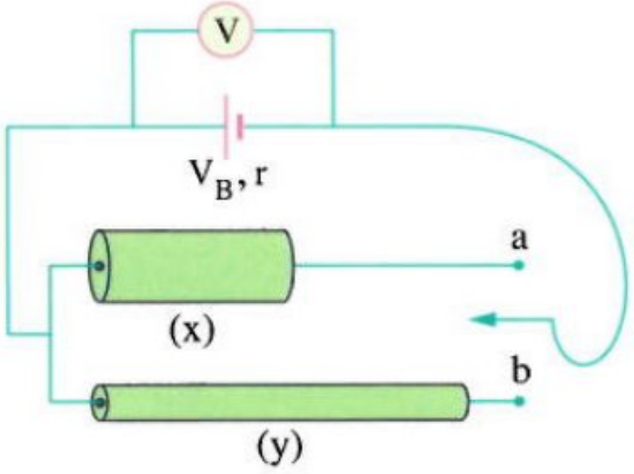


(a) Increases. (b) Decreases.

(c) Doesn't change. (d) The answer is indeterminable.

---

39 In the following figure, two conductors (x) and (y) of the same metal and uniform cross-sections have lengths (ℓ) and (2ℓ) and diameters (d) and $\left(\frac{d}{2}\right)$ respectively. They can be connected through a lead to a battery of internal resistance (r). And the following table records the voltmeter reading (V) as the lead changes its position of contact,



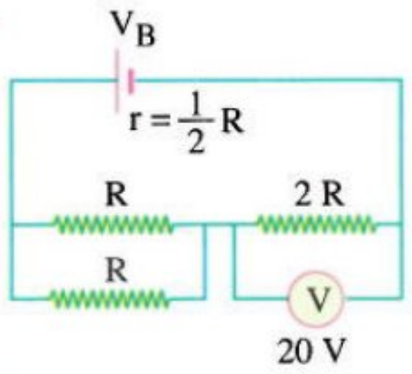
Case	Voltmeter reading
Not connecting the lead to either point (a) or (b)	V_1
Connecting the lead to point (a)	V_2
Connecting the lead to point (b)	V_3

Then, the correct order of the voltmeter readings V_1, V_2, V_3 is .....

(a) $V_1 = V_2 = V_3$ (b) $V_1 > V_2 > V_3$ (c) $V_1 > V_3 > V_2$ (d) $V_3 > V_2 > V_1$

---

40 In the opposite electric circuit, the magnitude of the electromotive force of the battery is ..... .

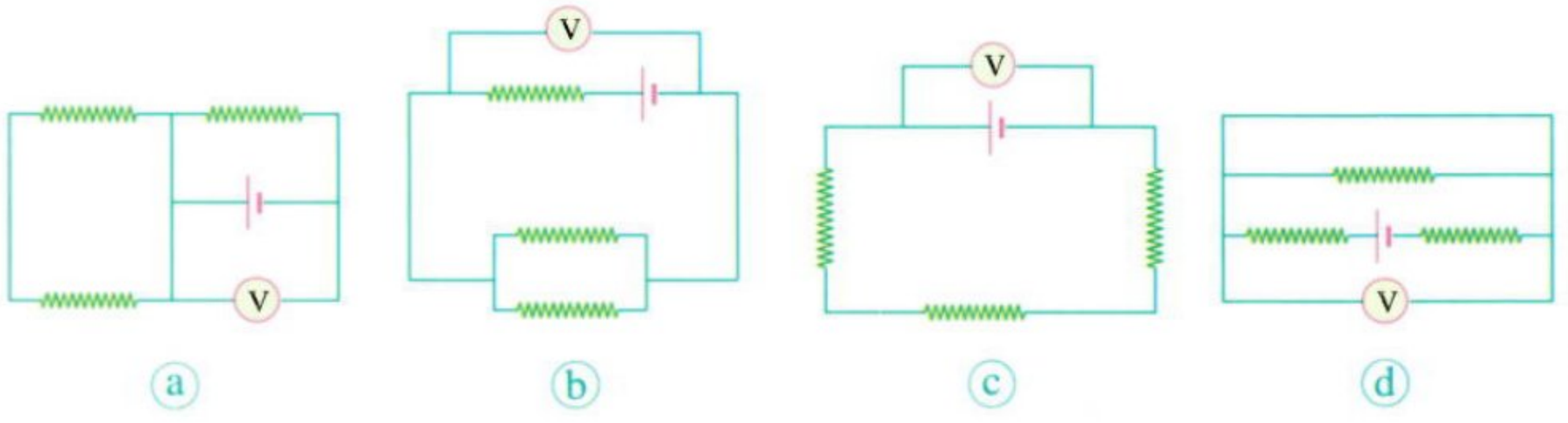


(a) 25 V (b) 30 V

(c) 35 V (d) 45 V

---

41 Three identical resistors are connected to a battery with internal resistance (r) in four different ways as shown in the following circuits. In which of these circuits will the voltmeter reading be the largest?

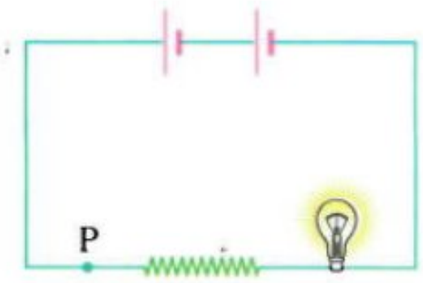


(a) (b) (c) (d)

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42 What is the change that could be done in the opposite circuit to increase the intensity of the passing electric current in the electric bulb?



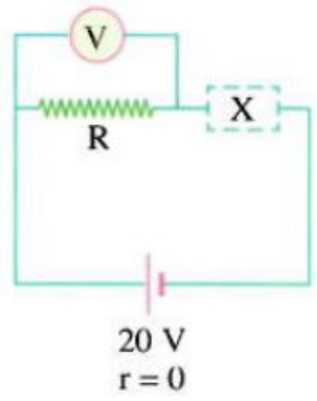
(a) Adding a resistor in parallel with the resistor which is in the circuit.

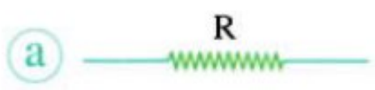
(b) Adding a resistor in series with the resistor which is in the circuit.

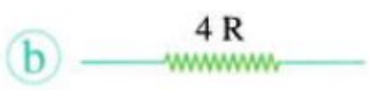
(c) Removing one of the two batteries.

(d) Moving the bulb to point P.

43 The opposite diagram represents a closed electric circuit, so which of the following figures represents the component X that makes the pointer of the voltmeter deflect to 4 V?



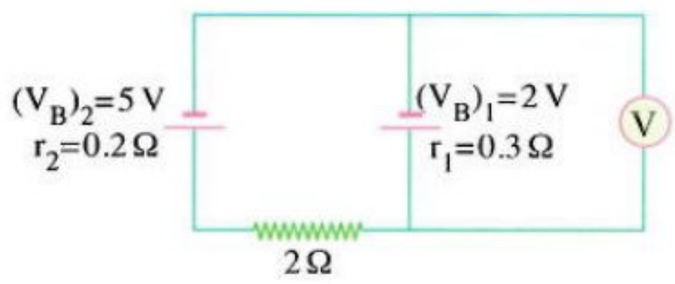
(a) 

(b) 

(c) 

(d) 

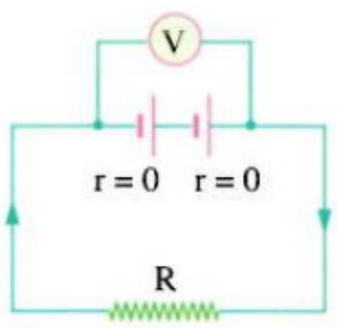
44 In the opposite electric circuit, the voltmeter reading is ..... .



(a) 7.64 V (b) 2.36 V

(c) 2 V (d) 1.64 V

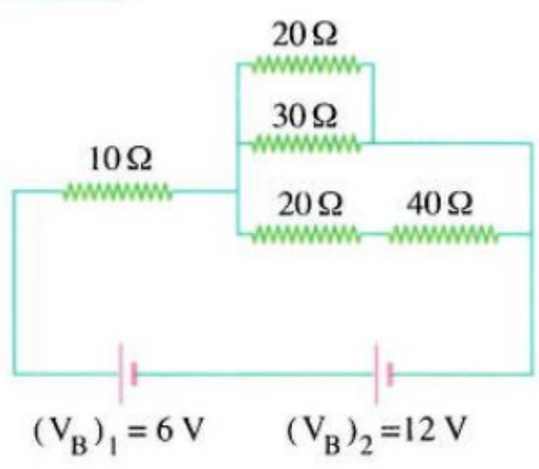
45 In the opposite electric circuit, if we remove one of the battery cells and connect another resistance in series to resistor R, the reading of the voltmeter ..... .



(a) increases (b) decreases

(c) remains unchanged (d) cannot be determined

46 ✳ In the opposite electric circuit, if the internal resistance of each electric cell is 2 Ω, then:



(i) The total external resistance of the circuit equals ..... .

(a) 5 Ω (b) 10 Ω

(c) 20 Ω (d) 25 Ω

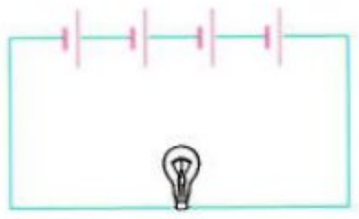
(ii) The total current intensity in the circuit equals ..... .

(a) 0.25 A (b) 0.5 A

(c) 0.75 A (d) 1 A

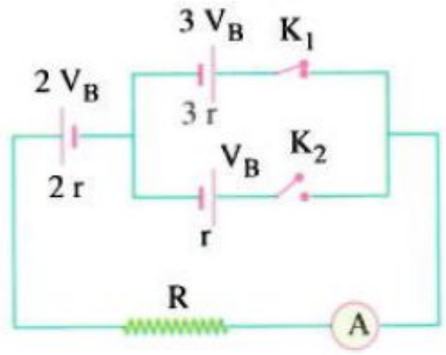
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47 ✳ The opposite figure shows four identical electric cells, each of emf 12 V, where they are connected with an electric bulb, when the intensity of the electric current that passes in the circuit was 0.5 A, the consumed electric power in the bulb was 23 W, so the internal resistance of each cell equals ..... .



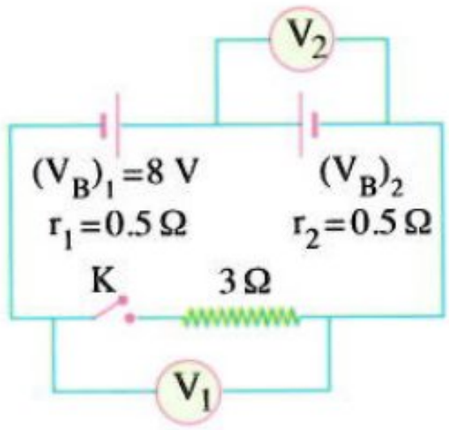
(a) 0.5 Ω (b) 1 Ω (c) 1.5 Ω (d) 2 Ω

48 ✳ In the shown electric circuit, when opening switch K_1 and closing switch K_2, which of the following choices shows what happen to the direction of the current in resistor R and the reading of the ammeter?



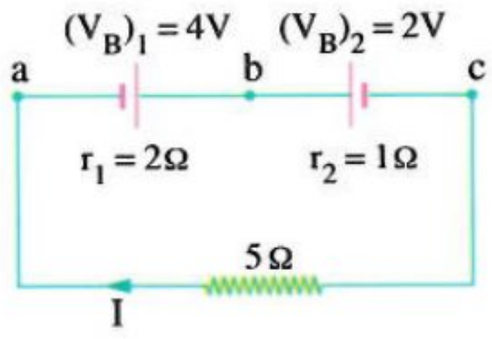
	Current direction	Ammeter reading
(a)	Remains unchanged	Increases
(b)	Gets reversed	Decreases
(c)	Remains unchanged	Decreases
(d)	Gets reversed	Increases

49 ✳ In the opposite electric circuit, if the reading of the voltmeter V_1 when the switch K is opened is 4 V, so if you know that $(V_B)_2 > (V_B)_1$, then the readings of each of the two voltmeters V_1 and V_2 after closing the switch K are ..... .



	The reading of V_1	The reading of V_2
(a)	3 V	11.5 V
(b)	3 V	8 V
(c)	4.5 V	11.5 V
(d)	4.5 V	8 V

50 ✳ In the opposite electric circuit:



(i) The current intensity passing in the circuit equals ..... .

(a) $\frac{2}{3}$ A (b) $\frac{3}{2}$ A

(c) $\frac{1}{2}$ A (d) $\frac{1}{4}$ A

(ii) The potential difference between the points a and b equals ..... .

(a) $\frac{7}{2}$ V (b) $\frac{5}{3}$ V (c) $\frac{4}{9}$ V (d) $\frac{2}{3}$ V

(iii) The potential difference between the points b and c equals ..... .

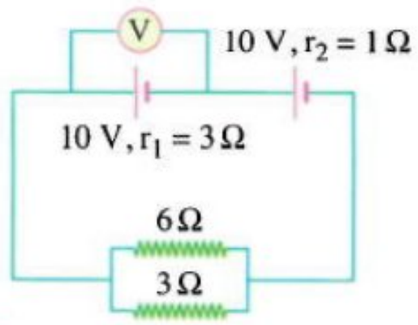
(a) 0.4 V (b) 2.25 V (c) 0.625 V (d) 1.6 V

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51 In the opposite electric circuit, the voltmeter reading is ..... .

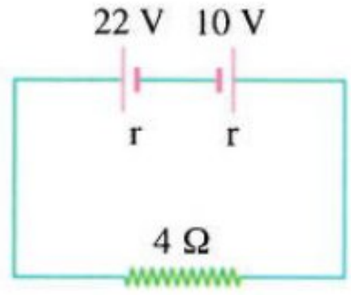


(a) 20 V (b) 10 V
(c) 5 V (d) 0

52 A rechargeable battery of electromotive force 8 V and internal resistance 0.4 Ω is connected to a source of internal resistance 0.4 Ω, so a current of intensity 2.5 A passes in the circuit. By ignoring the resistance of the wires in the circuit, the electromotive force of the source that is used to charge the battery is ..... .

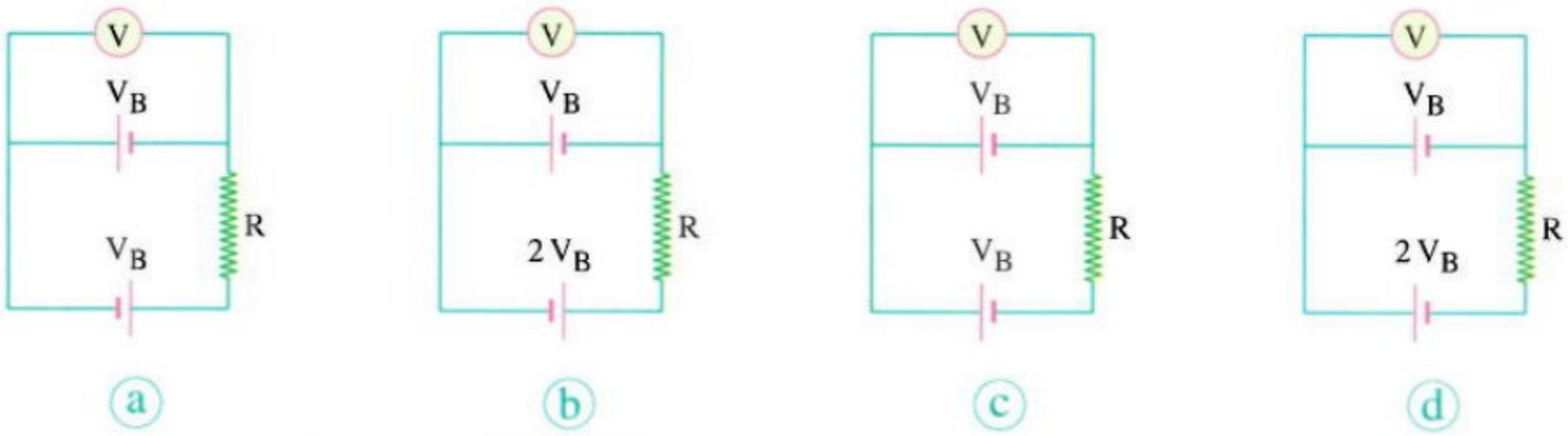
(a) 12 V (b) 11 V (c) 10 V (d) 6 V

53 In the opposite electric circuit, if the consumed power in the 4 Ω resistor is 16 W, then the value of r equals ..... .

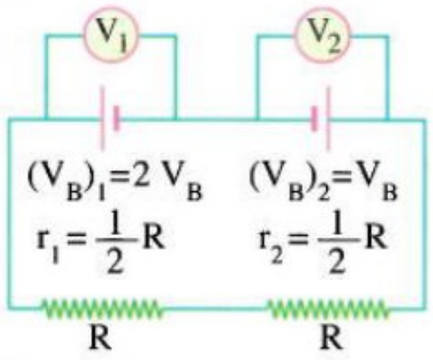


(a) 0.25 Ω (b) 0.5 Ω
(c) 1 Ω (d) 2 Ω

54 Each of the following electric circuits consists of a voltmeter, an ohmic resistance (R) and two batteries of the same internal resistance (r), so in which of these circuits is the voltmeter reading greater than electromotive force V_B?

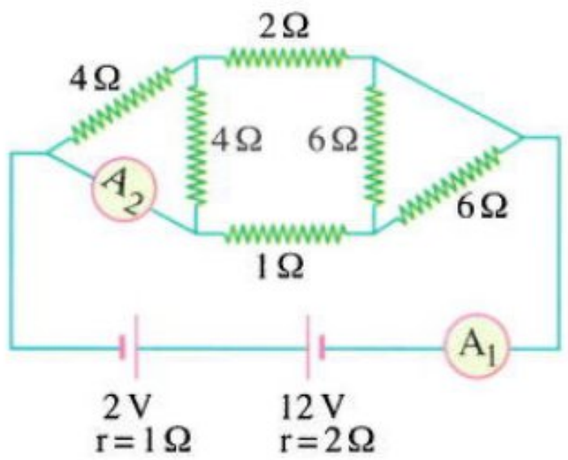


55 * In the opposite circuit, the ratio $\frac{V_2}{V_1}$ = ...



(a) $\frac{5}{11}$ (b) $\frac{2}{3}$
(c) $\frac{7}{11}$ (d) $\frac{1}{1}$

56 In the opposite electric circuit:



(i) The ammeter reading (A_1) equals ..... .

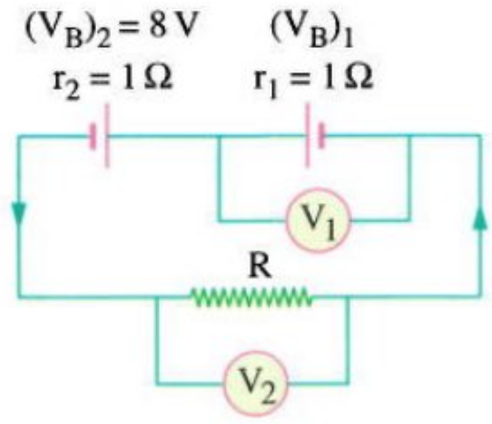
(a) 2 A (b) 5 A
(c) 7 A (d) 10 A

(ii) The ammeter reading (A_2) equals ..... .

(a) 0.5 A (b) 1 A
(c) 1.5 A (d) 2 A

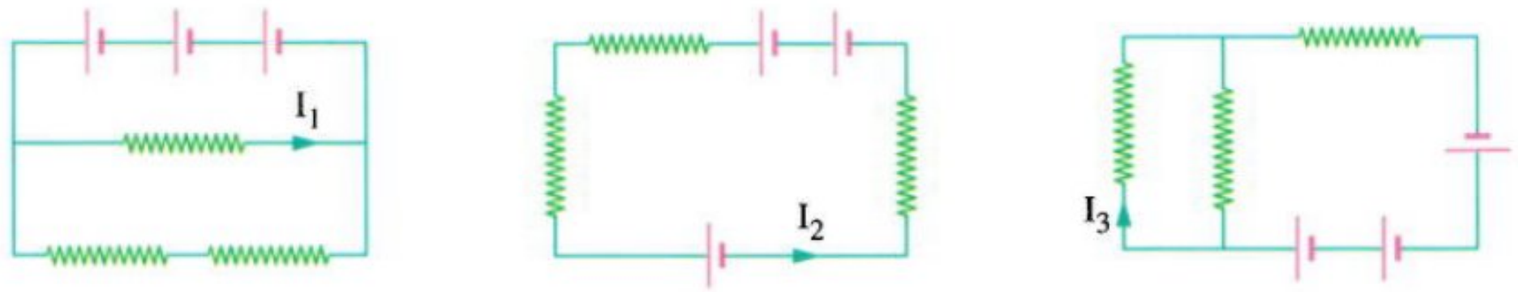
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57 In the opposite electrical circuit, if the voltmeter readings (V_1) and (V_2) are 16 V and 6 V respectively, then the value of resistance R and the electromotive force $(V_B)_1$ are respectively ....., ...



(a) 3 Ω, 16 V (b) 4 Ω, 10 V
(c) 3 Ω, 18 V (d) 4 Ω, 18 V

58 Three identical resistors are connected with three identical electric cells of negligible internal resistances in three different methods as follows:



(1) (2) (3)

So, the ratio of electric current intensities $I_1 : I_2 : I_3$ is .....

(a) 9 : 1 : 3 (b) 2 : 3 : 1 (c) 3 : 6 : 2 (d) 1 : 4 : 9

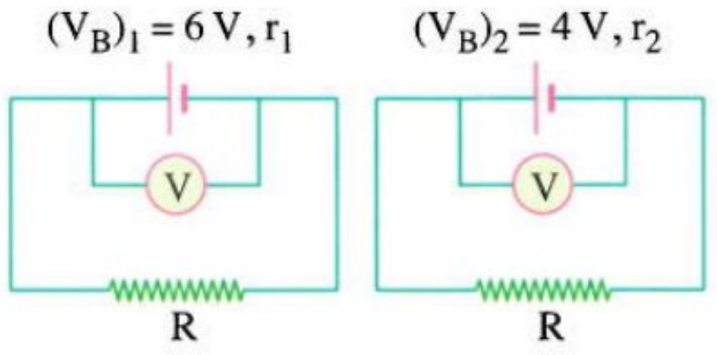
59 ✱ An electric cell was connected to a 1.9 Ω resistor, so an electric current of intensity 0.5 A passed in the circuit and when this resistor was replaced by another of 10.6 Ω, the current intensity decreased to 0.125 A, hence the emf of the cell equals ..... .

(a) 5.3 V (b) 3.31 V (c) 2.7 V (d) 1.45 V

60 ✱ An electric cell was connected to a resistor of resistance R, so the resistor R carried a current of intensity I and when another resistor of resistance $\frac{R}{2}$ was connected with the first resistor in parallel, the current intensity passing in the circuit was doubled, hence the internal resistance of the electric cell equals ..... .

(a) 3 R (b) $\frac{R}{3}$ (c) 6 R (d) $\frac{R}{6}$

61 A resistor of ohmic resistance R is connected to two different batteries, one at a time, as shown in the two opposite circuits. If the voltmeter readings in the two circuits are equal and equal to 3 V, then the ratio of the internal resistances of the two batteries $\left(\frac{r_1}{r_2}\right)$ equals ..... .



(a) $\frac{2}{3}$ (b) $\frac{3}{2}$
(c) $\frac{1}{3}$ (d) 3

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62 ✱ Two identical wires are made of the same material, each of 50 cm long and cross-sectional area of 2 mm^2. They are connected together in series with an electric cell of internal resistance 0.5 Ω to form an electric circuit, so the intensity of the passing current was 2 A and when the same two wires were connected in parallel with the same electric cell, the total electric current intensity became 6 A, so:

(i) The emf of the used electric cell equals ..... .

(a) 3 V (b) 6 V (c) 9 V (d) 12 V

(ii) The conductivity of the wire's material equals ..... .

(a) $125 \times 10^3\ \Omega^{-1}.m^{-1}$ (b) $250 \times 10^4\ \Omega^{-1}.m^{-1}$

(c) $396 \times 10^6\ \Omega^{-1}.m^{-1}$ (d) $431 \times 10^8\ \Omega^{-1}.m^{-1}$

63 ✱ A battery of emf 12 V is connected to two identical bulbs which are connected in parallel, so the potential difference between the terminals of the battery becomes 10.8 V. If each of the bulbs consumes an electric power of 12 W, then the internal resistance of the battery equals ..... .

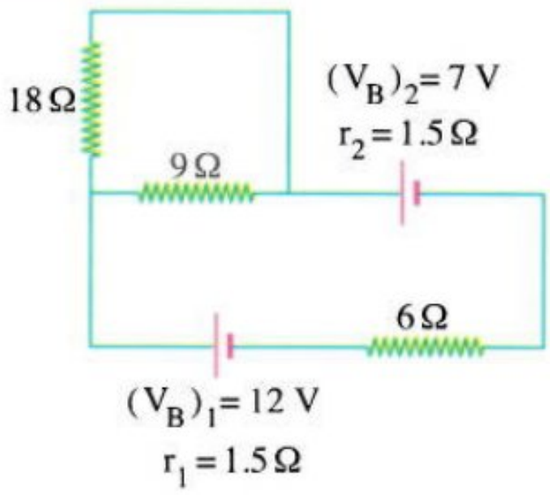
(a) 0.25 Ω (b) 0.54 Ω (c) 0.72 Ω (d) 1 Ω

64 ✱ From the opposite circuit:

(i) The current intensity passing through the 12 V battery equals ..... .

(a) $\frac{3}{5}$ A (b) $\frac{1}{3}$ A

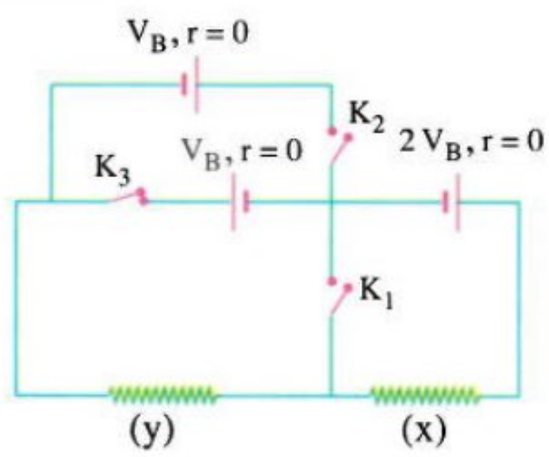
(c) $\frac{4}{9}$ A (d) $\frac{7}{8}$ A



(ii) The consumed power in the 9 Ω resistor equals ..... .

(a) $\frac{5}{4}$ W (b) $\frac{3}{7}$ W (c) $\frac{6}{5}$ W (d) $\frac{4}{9}$ W

65 Two identical resistors x and y are connected as shown in the opposite electric circuit, hence the electric power consumed in each of them is P_w, so what are the electric powers consumed in the two resistors x and y respectively when:



(i) closing switch K_1?

(a) $4 P_w, 2 P_w$ (b) $16 P_w, 4 P_w$

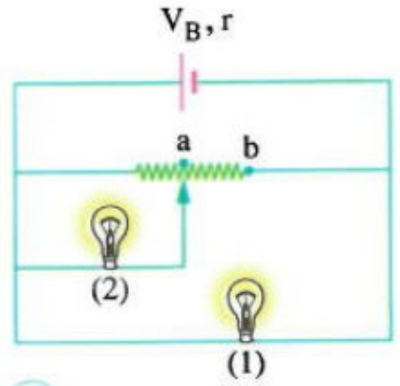
(c) $\frac{P_w}{4}, \frac{P_w}{2}$ (d) $\frac{P_w}{16}, \frac{P_w}{4}$

(ii) opening switch K_3 and closing switch K_2?

(a) $\frac{3}{2} P_w, 2 P_w$ (b) $\frac{P_w}{4}, \frac{P_w}{8}$ (c) $9 P_w, 9 P_w$ (d) $\frac{2 P_w}{3}, \frac{2 P_w}{3}$

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66 The opposite electric circuit contains two identical lightbulbs (1) and (2), so when moving the slider from position a to position b, what happens to the brightness of:



(i) bulb (1)?

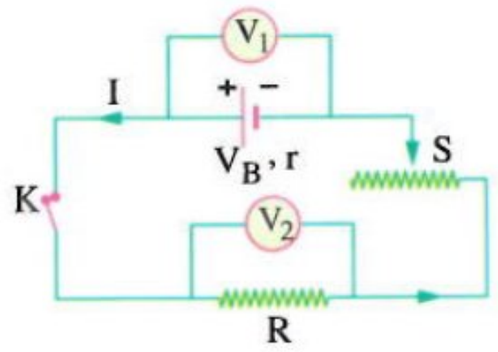
(a) Decreases. (b) Increases. (c) Doesn't change. (d) Vanishes.

(ii) bulb (2)?

(a) Decreases. (b) Increases. (c) Doesn't change. (d) Vanishes.

Second Essay questions

1 In the electric circuit shown in the figure:

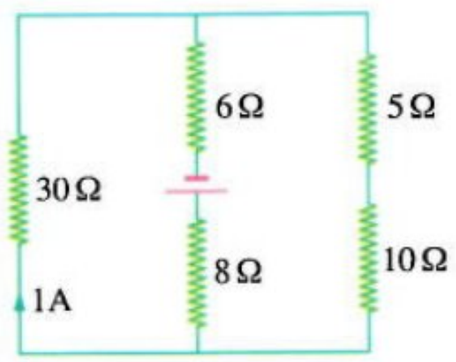


(a) **Write the relation between** the electric current intensity I passing in the circuit and the readings of V_1 and V_2

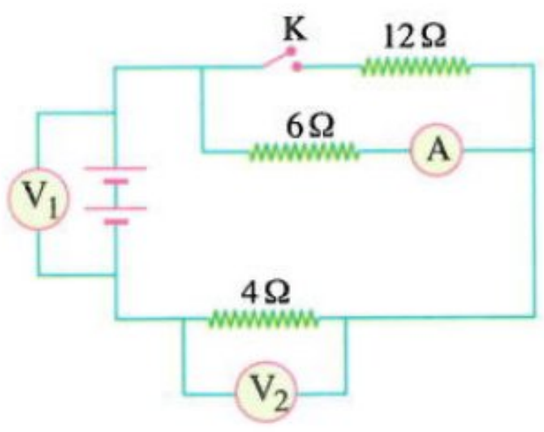
(b) **Deduce what happens to** the readings of each of V_1 and V_2 when the value of the rheostat resistance S increases.

(c) **What** is the reading of each of V_1 and V_2 when the switch K is opened?

2 From the opposite electric circuit, **calculate** the electromotive force of the source if its internal resistance equals 2 Ω.



3 In the opposite electric circuit, if the emf of the battery is 12 V and its internal resistance is 2 Ω, **find** the reading of each of the ammeter (A), the voltmeter (V_1), the voltmeter (V_2) when the switch K is:



(a) opened. (b) closed.

4 In the two opposite figures (1) and (2), a variable resistance R is connected once to an electric source of an electromotive force V_B and neglected internal resistance and again connected to a source of electromotive force V_B and internal resistance r, when the resistance R increases in the two circuits the voltmeter reading V_1 remains constant, while the voltmeter reading V_2 increases. **Explain**.

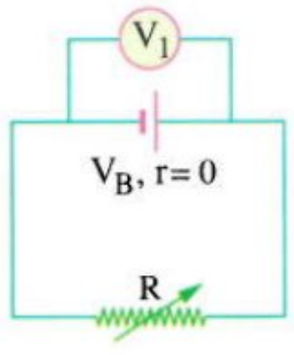


Figure (1)

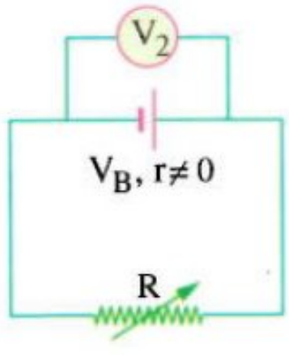


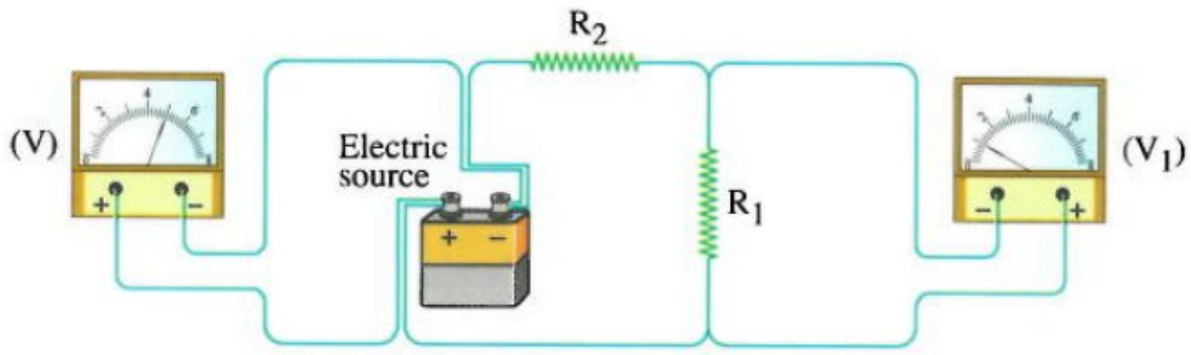
Figure (2)

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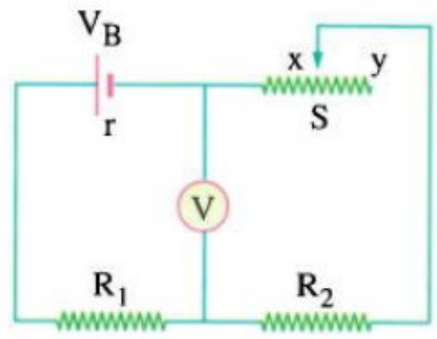
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5 In the following electric circuit, if resistor R_1 is equivalent to double the internal resistance (r) of the source and the reading of the voltmeter (V) across the source is five times as the reading of the voltmeter (V_1) across R_1,



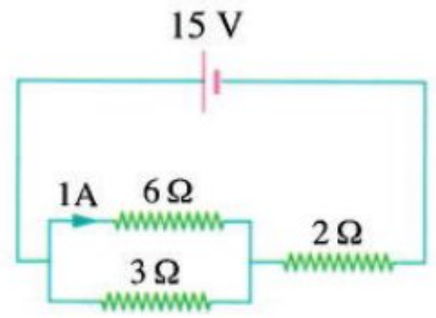
find the resistance of R_2 in terms of the internal resistance of the source (r).

6 In the opposite circuit, **what happens to** the voltmeter reading when the slider moves from point x to point y?

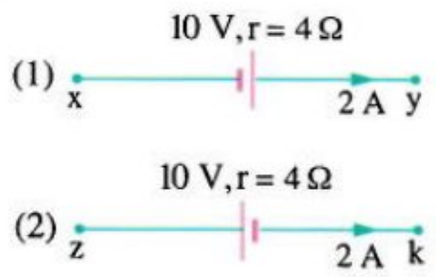


7 An electric circuit consists of a battery whose electromotive force is V_B, internal resistance r, variable resistance R and an ammeter, when the resistance R is doubled, the reading of the ammeter doesn't decrease to its half. **Does this agree with Ohm's law? Explain.**

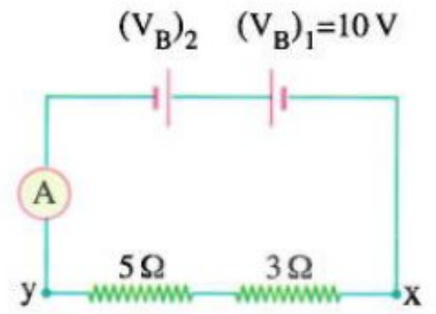
8 In the opposite circuit, **is** the internal resistance of the battery considered or neglected ? **Explain your answer.**



9 The two figures (1) and (2), each one represents a part of a closed electric circuit, **find** the ratio between the potential differences $\left(\frac{V_{xy}}{V_{zk}}\right)$.

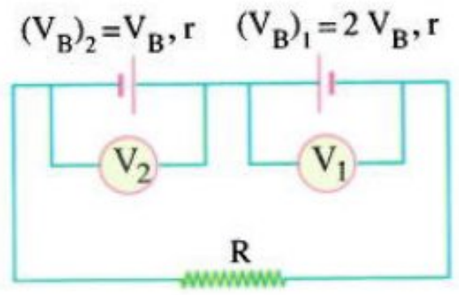


10 In the opposite electric circuit, if the electric potential of point x is greater than the electric potential of point y and the ammeter reading is 2 A and the internal resistance of each battery is 1 Ω. **Calculate** the electromotive force of the battery $(V_B)_2$



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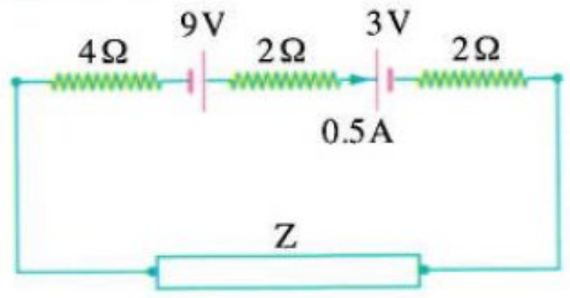
11 **From the opposite figure, which is greater with an explanation:**



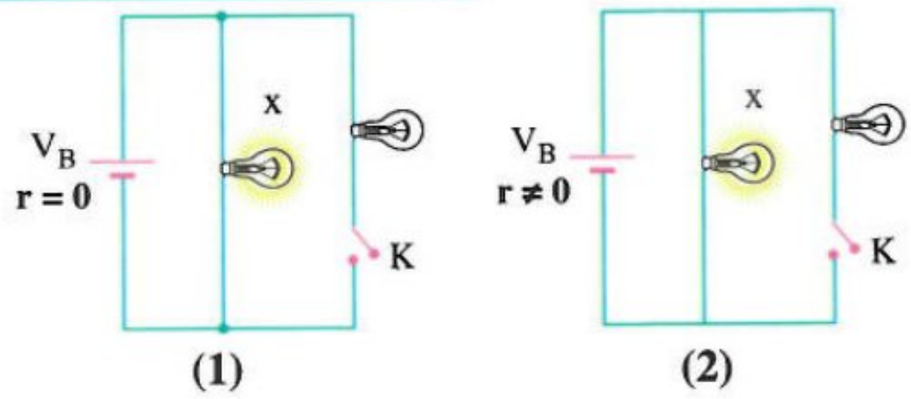
(a) $(V_B)_1$ or V_1?

(b) $(V_B)_2$ or V_2?

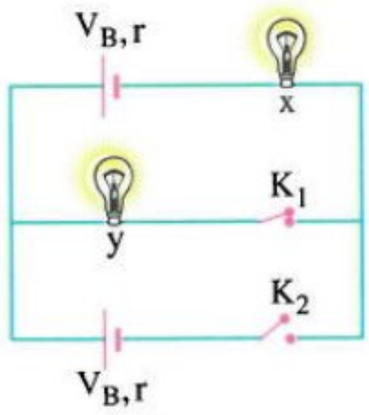
12 The opposite figure shows a closed electric circuit containing an unknown component Z. Considering the internal resistances of the electric sources are negligible, **present two options** for the unknown component Z.



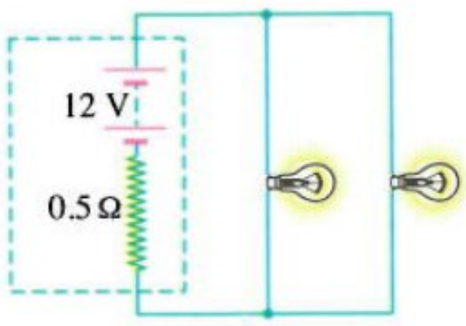
13 From the opposite two circuits (1) and (2), **what is the effect of** closing the switch on the brightness of lamp x in each of the two circuits?



14 In the opposite electric circuit, **what happens to** the brightness of lamp x when the switch K_1 is opened and switch K_2 is closed?



15 Two identical electric lamps rated (12 V, 60 W) are connected to a battery whose electromotive force is 12 V and internal resistance is 0.5 Ω as shown in the opposite figure, **why** is the power consumed by each lamp not 60 W?

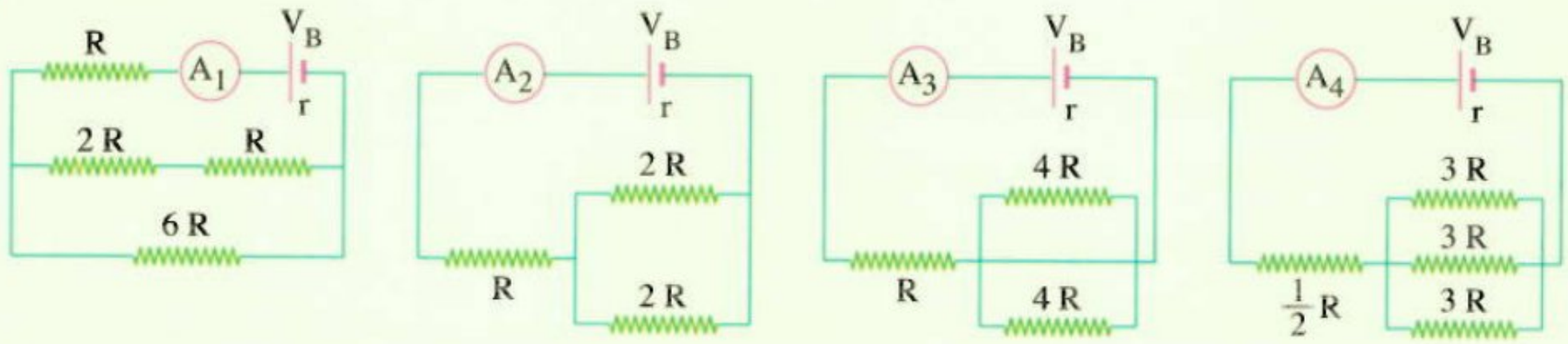


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Questions of the Previous Years Exams on Lesson 3

1



Each of the four electric circuits contains an ammeter. What is the correct arrangement of the readings A_1, A_2, A_3 and A_4 of the four ammeters? *(Exp. May 21)*

(a) $A_4 < A_2 < A_1 < A_3$ (b) $A_4 < A_3 < A_1 < A_2$

(c) $A_3 < A_4 < A_2 < A_1$ (d) $A_1 < A_2 < A_4 < A_3$

2 An electric cell of unknown emf is connected to resistance R_1, hence the intensity of the passing current through the cell equals 0.5 A as in figure (1) and when replacing resistance R_1 by another resistance R_2, the intensity of the passing current through the cell becomes 0.3 A as in figure (2), so the emf of the cell equals ..... . *(Exp. May 21)*

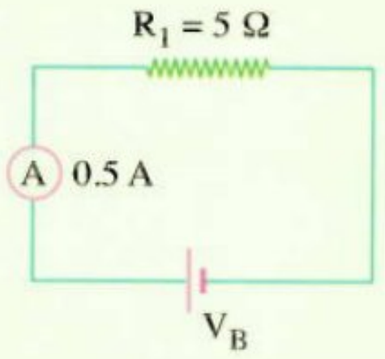


Figure (1)

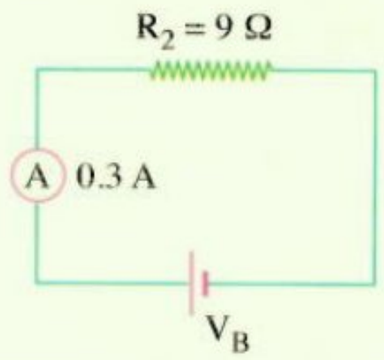


Figure (2)

(a) 1.2 V (b) 1.5 V (c) 2 V (d) 3 V

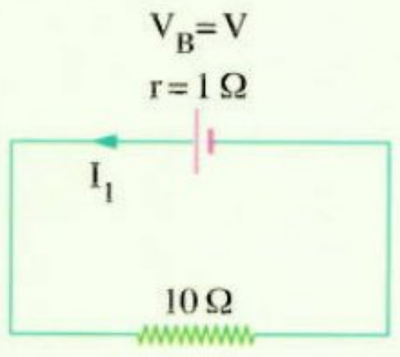
3 In the opposite figure, the ratio $\left(\frac{I_1}{I_2}\right)$ equals ..... . *(Exp. June 21)*

(a) $\frac{6}{11}$

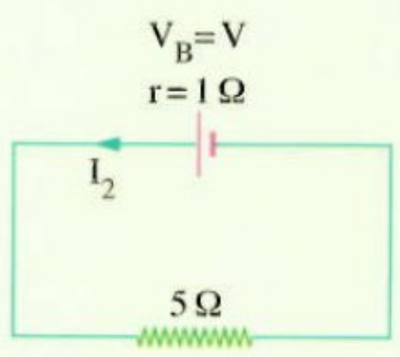
(b) $\frac{11}{6}$

(c) $\frac{1}{2}$

(d) $\frac{1}{1}$



Circuit (1)

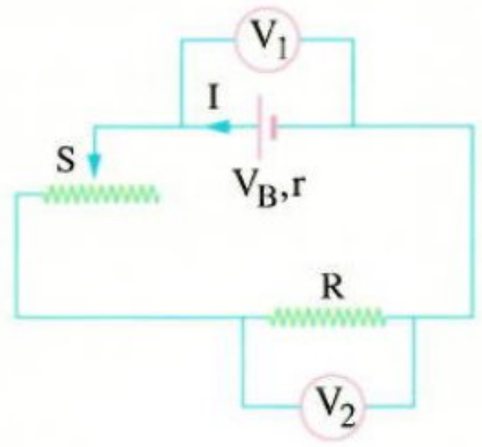


Circuit (2)

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4 In the opposite closed electric circuit, the ratio

of $\frac{\text{the reading of voltmeter } V_1}{\text{the reading of voltmeter } V_2}$ = ... *(1st session 21)*

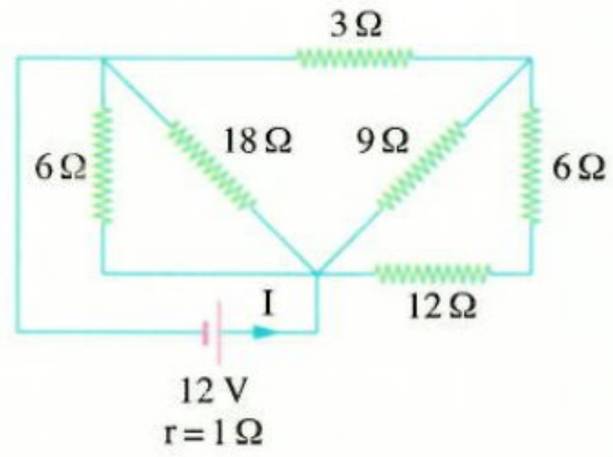


(a) $\frac{IR}{V_B + V_2}$ (b) $\frac{IR - Ir}{V_2 - V_B}$

(c) $\frac{V_B - Ir}{IR}$ (d) $\frac{V_B + Ir}{IR}$

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5 In the shown electric circuit, the electric current intensity (I) equals ..... . *(1st session 21)*



(a) 4 A

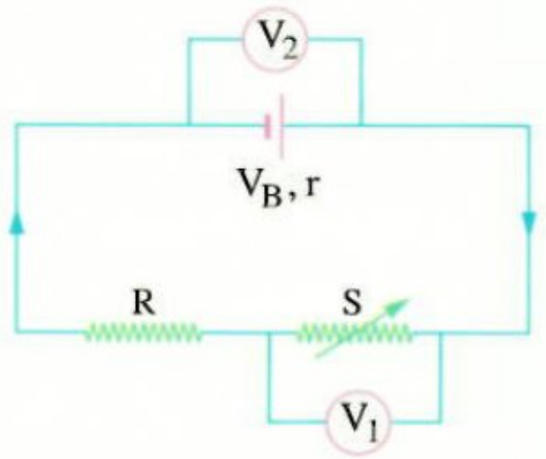
(b) 0.76 A

(c) 0.83 A

(d) 3 A

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6 In the opposite closed electric circuit, when the value of resistance S is increased, ... *(2nd session 21)*



(a) the readings of both (V_1) and (V_2) increase

(b) the reading of (V_1) increases while of (V_2) decreases

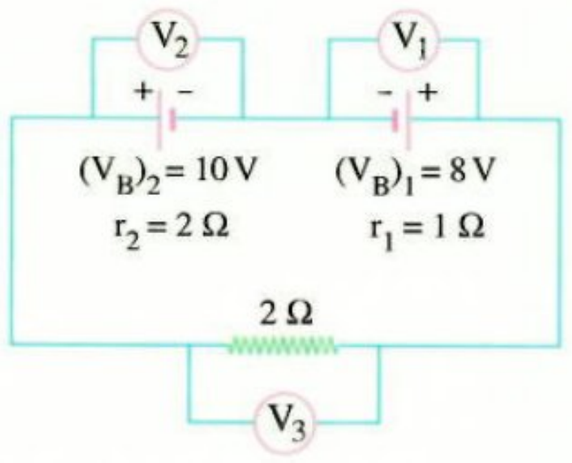
(c) the reading of (V_1) decreases while of (V_2) increases

(d) the readings of both (V_1) and (V_2) decrease

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7 In the opposite electric circuit: If the reading of voltmeter V_3 equals 0.8 V. Which choice correctly represents the readings of V_1 and V_2?

(2nd session 21)



	V_1	V_2
(a)	10 V	6 V
(b)	8.4 V	9.2 V
(c)	7.6 V	9.2 V
(d)	4 V	8 V

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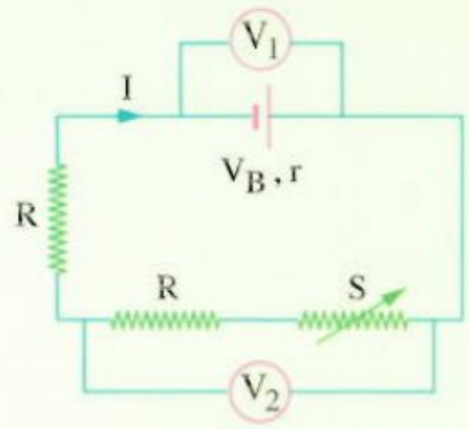
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8 In the shown electric circuit:

On increasing the value of the variable resistance (S).

Which choice correctly represents the change in the readings of voltmeters (V_1) and (V_2)? *(1st session 22)*

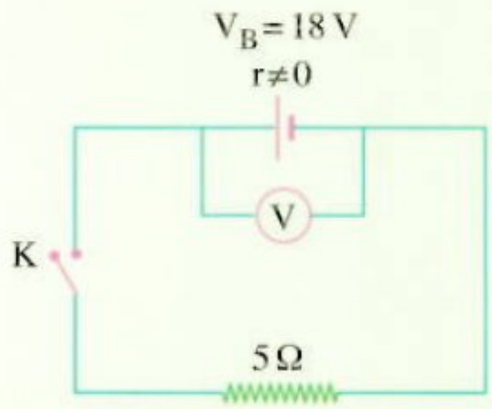


	V_1	V_2
a	Increases	Increases
b	Remains constant	Increases
c	Decreases	Remains constant
d	Decreases	Decreases

9 If the voltmeter reads 18 V when the switch (K) is opened, but reads 15 V when the switch is closed, so the internal resistance of the cell = ... *(1st session 22)*

a) 3 Ω b) 2 Ω

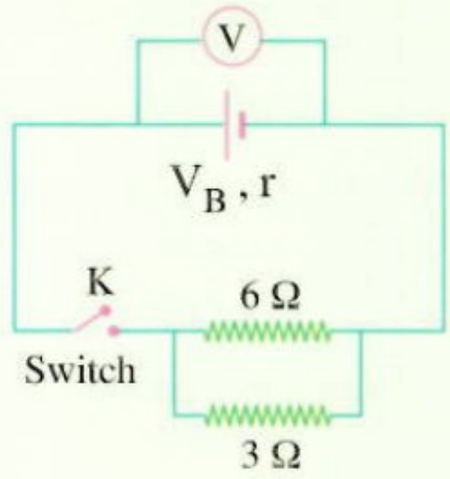
c) 4 Ω d) 1 Ω



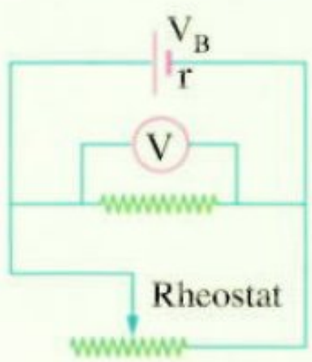
10 In the electric circuit shown, the voltmeter reads 14 V when the switch (K) is opened, but reads 8 V when the switch (K) is closed. Then the internal resistance of the battery = ... *(2nd session 22)*

a) 1.25 Ω b) 0.5 Ω

c) 1.5 Ω d) 0.25 Ω



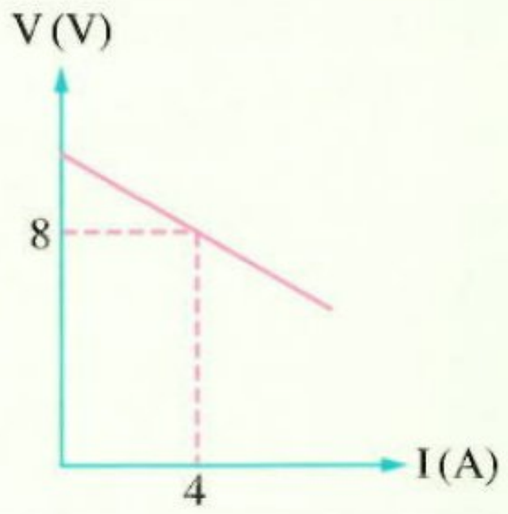
11 According to the opposite circuit, which of the following represents what will happen to the reading of the voltmeter by changing the value of resistance which is taken from rheostat? *(Exp. 23)*



	The value of the resistance taken from the rheostat	The reading of the voltmeter
a	Decreases	Decreases
b	Decreases	Increases
c	Increases	Decreases
d	Increases	Doesn't change

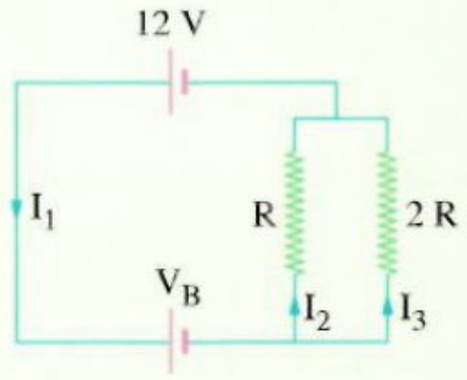
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12 The opposite graph shows the relation between the terminal voltage of a battery (V) that has an internal resistance 0.5 Ω connected with a closed electric circuit and the current intensity (I) that passes through it, then the value of the electromotive force of the battery is ..... . *(Exp. 23)*



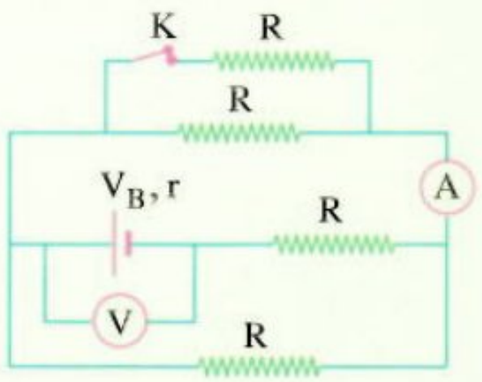
(a) 8 V (b) 10 V
(c) 9 V (d) 12 V

13 As in the shown circuit, which of the following is correct with respect to the value of V_B, I_1 and I_2? *(Exp. 23)*



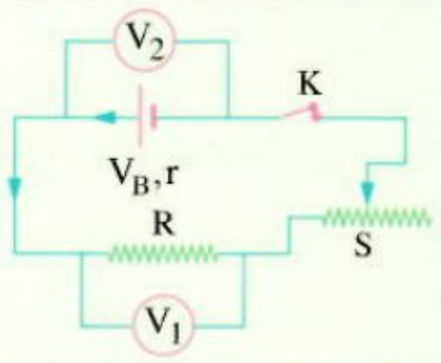
	V_B	I_1	I_2
(a)	6 V	2 A	1 A
(b)	18 V	3 A	1 A
(c)	18 V	1 A	2 A
(d)	6 V	3 A	2 A

14 The figure illustrates a closed electric circuit. So, when the switch (K) is opened, ... *(1st session 23)*



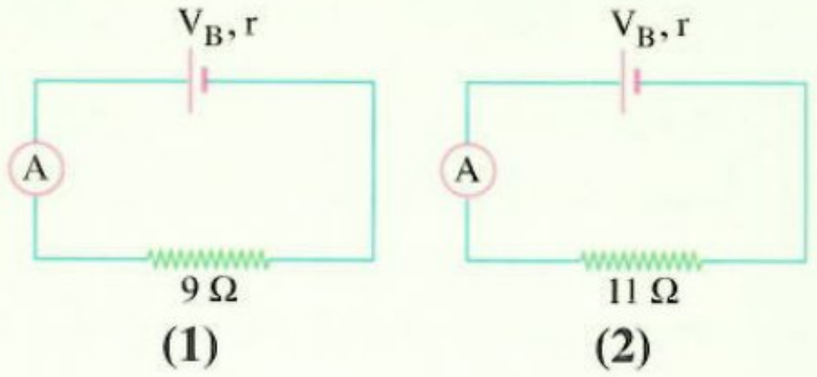
(a) the ammeter reading decreases, while the voltmeter reading increases
(b) the ammeter reading increases, while the voltmeter reading decreases
(c) the reading of both ammeter and voltmeter decreases
(d) the reading of both ammeter and voltmeter increases

15 In the shown closed electric circuit:
We can conclude that ..... . *(2nd session 23)*



(a) $V_2 < V_B$ (b) $V_1 > V_B$
(c) $V_2 = V_B$ (d) $V_1 = V_2$

16 The figure illustrates a cell with internal resistance (r), when it is connected to 2 different circuits (each individually), the ammeter reading is (1.2 A) in the first circuit and (1 A) in the second circuit. So the value of the internal resistance of the cell = ... *(2nd session 23)*



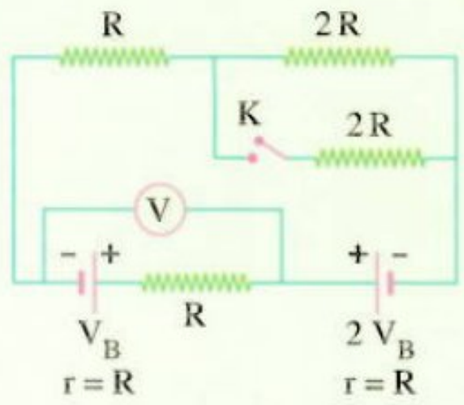
(a) 1.5 Ω (b) 2 Ω (c) 0.5 Ω (d) 1 Ω

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17 In the shown electric circuit, which choice is correct?

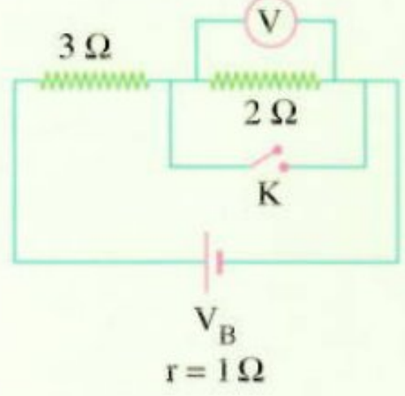
	Reading of voltmeter when K is opened	Reading of voltmeter when K is closed
(a)	$\frac{4}{3}V_B$	$\frac{6}{5}V_B$
(b)	$\frac{4}{3}V_B$	$\frac{7}{5}V_B$
(c)	$\frac{7}{6}V_B$	$\frac{6}{5}V_B$
(d)	$\frac{7}{6}V_B$	$\frac{7}{5}V_B$



(1st session 24)

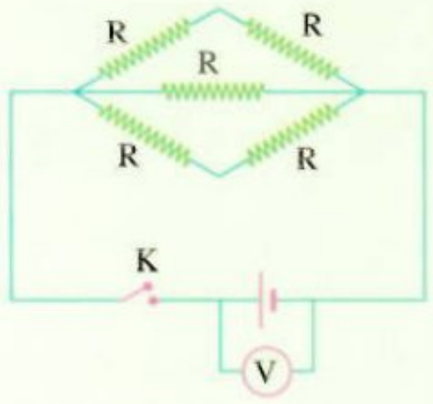
18 The opposite figure represents an electric circuit, if the reading of the voltmeter, when the key is open = 4 V Then the potential difference between the terminals of the 3 Ω resistor when the key is closed = ... *(2nd session 24)*

(a) 4 V (b) 8 V
(c) 6 V (d) 9 V



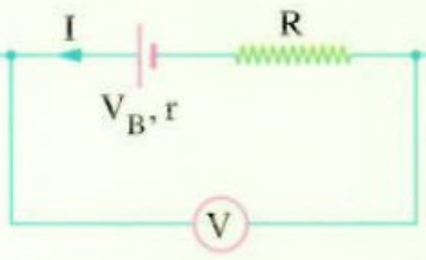
19 In the opposite electric circuit, if the internal resistance of the battery is 0.5 Ω and the reading of the voltmeter when switch K is open is 21 V and when it is closed, is 19.5 V, then the intensity of the electric current in the circuit and the value of resistance R respectively are ....., ... *(Practice 25)*

(a) 2 A, 10 Ω (b) 2 A, 13 Ω (c) 3 A, 10 Ω (d) 3 A, 13 Ω



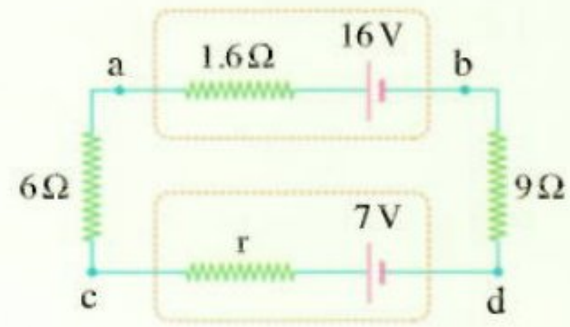
20 The opposite figure represents section of electric circuit, so the reading of the voltmeter (V) can be calculated from the relation; ..... *(Practice 25)*

(a) $V = V_B - I(R + r)$ (b) $V = V_B - I(R - r)$
(c) $V = V_B + I(R + r)$ (d) $V = V_B + I(R - r)$



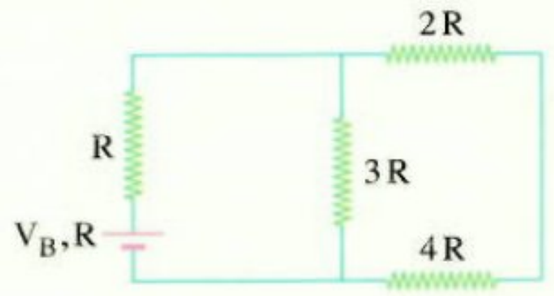
21 In the electrical circuit shown in the opposite figure, if the potential difference between points a and b is 15.2 V, then the value of r and the potential difference between points c and d respectively are ....., ... *(Practice 25)*

(a) 1.4 Ω, 7.7 V (b) 1.4 Ω, 7.8 V (c) 0.7 Ω, 7.7 V (d) 2.8 Ω, 8.4 V



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22 In the opposite electrical circuit, a group of different resistors are connected together with a battery of internal resistance R. Which of the following relationships for the potential differences across the external resistors is correct ? *(Practice 25)*



(a) $V_{3R} = 3\ V_R = 3\ V_{2R} = 1.5\ V_{4R}$

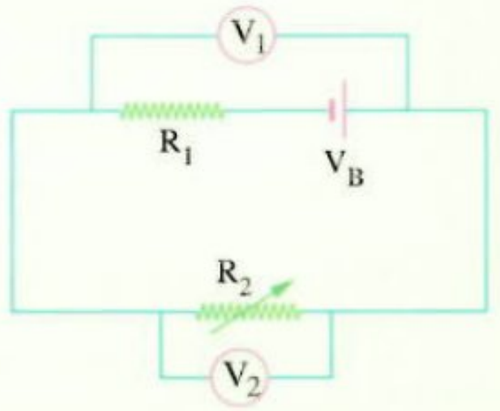
(b) $V_{3R} = 3\ V_R = 1.5\ V_{2R} = 0.75\ V_{4R}$

(c) $V_{3R} = 2\ V_R = V_{2R} = V_{4R}$

(d) $V_{3R} = 2\ V_R = 3\ V_{2R} = 1.5\ V_{4R}$

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23 In the electrical circuit shown in the figure, what happens to the readings of the voltmeters (V_1) and (V_2) when the value of the variable resistance (R_2) is increased?

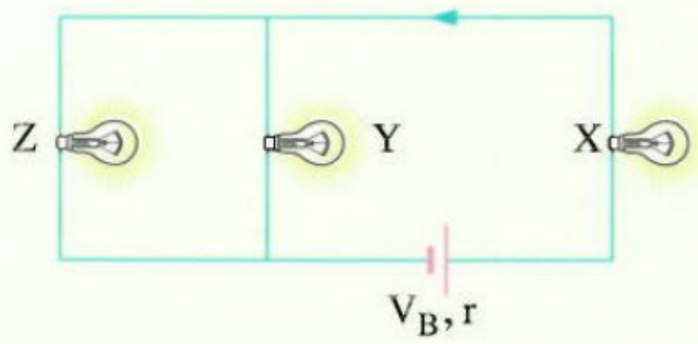


	Voltmeter reading (V_1)	Voltmeter reading (V_2)
(a)	Decreases	Decreases
(b)	Decreases	Increases
(c)	Increases	Decreases
(d)	Increases	Increases

(1st session 25)

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24 Three identical lamps, each is of resistance (R), are connected to a battery of electromotive force (V_B) and internal resistance equal to (0.5 R) as shown in the figure. If the filament of lamp (Y) is damaged, what happens to the glowing intensity (brightness) of the two lamps (X) and (Z)?



	Brightness of lamp (X)	Brightness of lamp (Z)
(a)	Increases	Increases
(b)	Increases	Decreases
(c)	Decreases	Increases
(d)	Decreases	Decreases

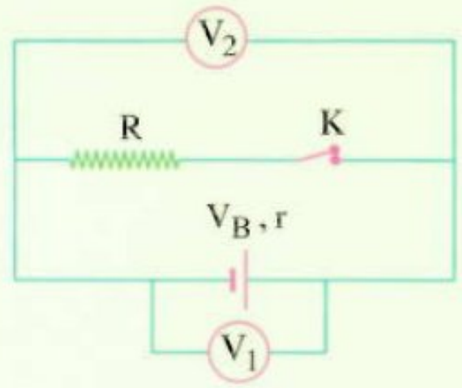
(1st session 25)

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25 The electrical circuit shown in the figure contains a resistance (R) connected to a battery of electromotive force (V_B) and internal resistance (r).



What will happen to the readings of the voltmeters V_1 and V_2 when switch (K) is opened?

	Voltmeter reading (V_1)	**Voltmeter reading (V_2)**
a	Increases to V_B	Increases to V_B
b	Increases to V_B	Decreases to zero
c	Decreases to zero	Increases to V_B
d	Decreases to zero	Decreases to zero

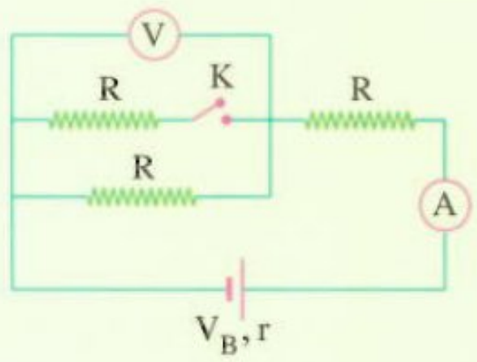
(2nd session 25)

26 The diagram shows a 20 Ω resistance connected across the terminals of a battery having an electromotive force of 12 V and internal resistance of 4 Ω. If a voltmeter of resistance 80 Ω is connected in parallel across this resistor, what is the voltmeter reading? *(2nd session 25)*



a) 12 V b) 10 V c) 9.6 V d) 2.4 V

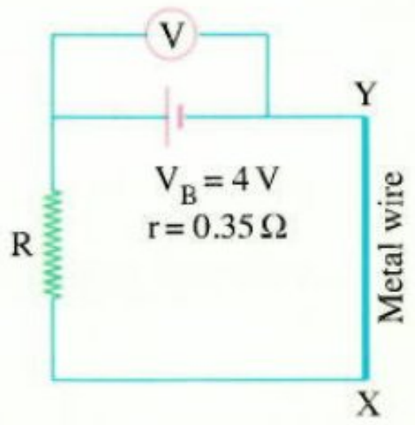
27 In the electric circuit shown in the figure, the ammeter reading is (I), the voltmeter reading is (V), and the internal resistance of the battery equals (r = 0.5 R). What will happen to the ammeter and voltmeter readings when the switch (K) is closed? *(2nd session 25)*



	The ammeter reading	**The voltmeter reading**
a	Increases	Decreases
b	Increases	Increases
c	Decreases	Decreases
d	Decreases	Increases

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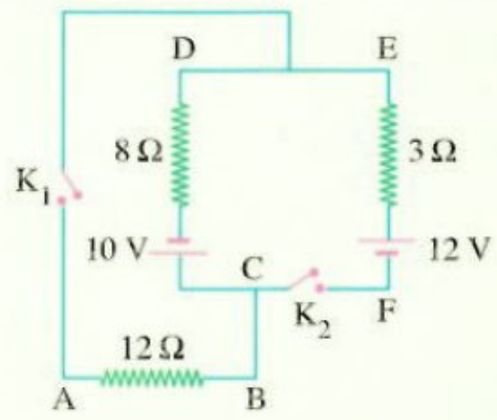
28 The figure shows an electric circuit consisting of a battery of electromotive force 4 V and of an internal resistance 0.35 Ω connected to a fixed resistance R and a metal wire XY of length 90 cm and cross-sectional area 3×10^{-8} m^2. If the reading of the voltmeter is 3.3 V, while the potential difference between the two points X and Y is 1.8 V, then :



(Practice 26)

	The value of the equivalent external resistance	The electric conductivity of wire XY material
(a)	1.65 Ω	33.33×10^6 $\Omega^{-1}.m^{-1}$
(b)	0.9 Ω	33.33×10^6 $\Omega^{-1}.m^{-1}$
(c)	1 Ω	3×10^8 $\Omega^{-1}.m^{-1}$
(d)	0.75 Ω	3×10^8 $\Omega^{-1}.m^{-1}$

29 Using the given data on the opposite electric circuit, when only switch K_1 is closed, the potential difference between points B and D is equal to ....., while when only switch K_2 is closed, the potential difference between points D and F is equal to ..... .



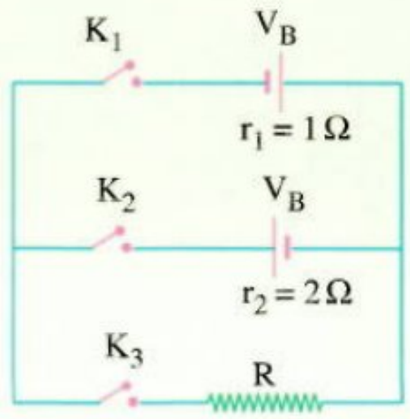
(Practice 26)

	Potential difference between points B and D at closing K_1 only	Potential difference between points D and F at closing K_2 only
(a)	14 V	6 V
(b)	6 V	16 V
(c)	6 V	6 V
(d)	26 V	18 V

30 In the shown electric circuit when closing K_1 and K_3 only, a current of 0.8 A passes. And when closing K_2 and K_3 only, a current of 0.6 A passes, **calculate** the value of V_B *(1st session 24)*

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A battery of electromotive force 18 V and internal resistance 2 ohms is connected with resistance R, if the potential difference between the two terminals of the battery is 12 V. **Calculate** the electric current intensity in the circuit if a resistance of 12 ohms is connected in parallel with the resistance R. *(2nd session 24)*

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